FIITJEE **JEE (Main)-2018**

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST - III

Time Allotted: 3 Hours **Maximum Marks: 360**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section-A (01 – 30, 31 – 60, 61 – 90)** *contains 90 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

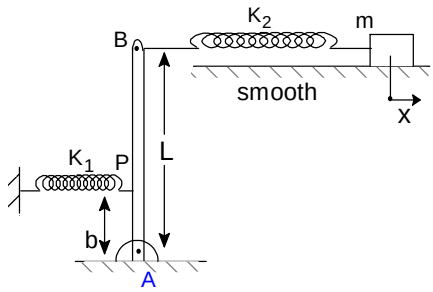
Physics

PART – I

SECTION – A
(One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A massless Rod AB is hinged at point A. At point P, it is connected by a spring of stiffness K_1 and at point B it is connected to another spring of stiffness K_2 connected to mass m as shown in figure. The length of Rod AB is ℓ and point P is at a distance b from point A. if mass m is displaced by a distance x from its mean position as shown in figure then the time period of S.H.M. of mass is



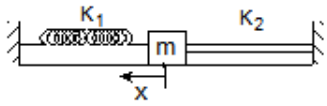
(A) $2\pi\sqrt{\dfrac{m\left(K_2b^2+K_1L^2\right)}{K_1K_2b^2}}$

(B) $2\pi\sqrt{\dfrac{m\left(K_2L^2+K_1b^2\right)}{K_1K_2b^2}}$

(C) $2\pi\sqrt{\dfrac{m\left(K_1K_2b^2\right)}{K_2b^2+K_1L^2}}$

(D) $2\pi\sqrt{\dfrac{m\left(K_2L^2+K_1b^2\right)}{K_2b^2}}$

2. A block of mass m is connected to an ideal spring of stiffness K_1 on left hand side and to an elastic string of stiffness K_2 on right hand side as shown in figure. The time period of motion if mass m is displaced by 'x' as shown in figure.



(A) $2\pi\sqrt{m/K_1}$

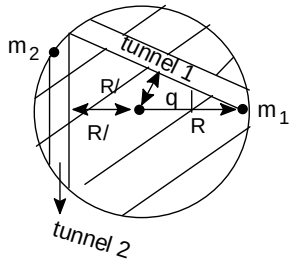
(B) $2\pi\sqrt{m/\left(K_1+K_2\right)}$

(C) $2\pi\sqrt{m\left(K_1+K_2\right)/K_1K_2}$

(D) $\pi\left(\sqrt{\dfrac{m}{K_1+K_2}}+\sqrt{\dfrac{m}{K_1}}\right)$

Space for Rough work

3. Two mass m_1 & m_2 are initially placed at rest inside two smooth tunnels in earth as shown in the figure. The minimum distance between the tunnels and the center of earth is R/2 as shown in the figure. Then



(A) m_1 will remain at rest and m_2 will perform SHM with

$$T.P._2 = 2\pi\sqrt{\frac{R}{2g}}$$

(B) m_1 will remain at rest and m_2 will perform SHM with

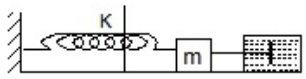
$$T.P._2 = 2\pi\sqrt{\frac{R}{g}}$$

(C) m_1 & m_2 both will perform S.H.M with same time period of

$$2\pi\sqrt{\frac{R}{g}}$$

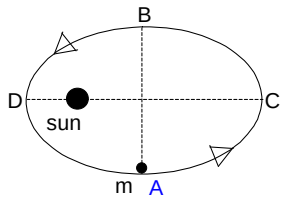
(D) m_1 will remain at rest and m_2 will move down to other side of tunnel then move outside the earth tunnel.

4. For a damped oscillator which follow the equation $\vec{F} = -K\vec{x} - b\vec{V}$ the mass of the block is 100gm K = 100 N / M and the damping constant is 20 gm / sec. Then find the time taken for its mechanical energy to drop to one – fourth of its initial value.



(A) 1.73 (B) 3.46
(C) 6.93 (D) 13.84

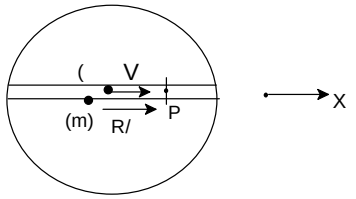
5. A planet of mass m moves in an elliptical orbit around sun with sun at its Focus. (e = eccentricity of ellipse). Then T_{ACB} / T_{BDA} is [T_{ACB} is time taken to travel from A to C to B & T_{BDA} is time taken to travel from B to D to A]



(A) $(\pi + 2e)/(\pi - 2e)$ (B) 1
(C) $(\pi + e)/(\pi - e)$ (D) $\pi/(\pi - 1)$

Space for Rough work

6. An object of mass is projected from center of earth along a smooth tunnel dug along the horizontal diameter as shown in the figure. Then choose the condition on values of v for the object to cross P at least twice.



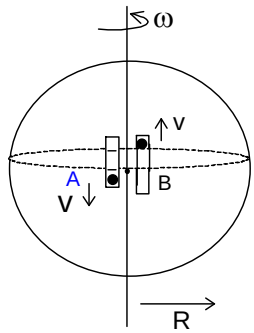
(A) $\frac{\sqrt{gR}}{2} < V < \sqrt{gR}$

(B) $\left(\sqrt{gR}\right) < V < \sqrt{3gR}$

(C) $\sqrt{gR}\ /\ 2 < V < \sqrt{3gR}$

(D) $\sqrt{gR}\ /\ 4 < V < \sqrt{3gR}$

7. Two trains A and B each of mass m are moving on a longitude of a spherical planet of mass M and radius R with velocity v relative to the planet. Train A & B presses the rail track with forces N_A & N_B respectively. [ω is angular velocity of planet about its vertical axis]Then



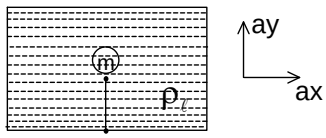
(A) $N_A = N_B = \frac{GmM}{R^2} - m\omega^2 R$

(B) $N_A = \frac{GmM}{R^2} - m\omega^2 R + \frac{mv^2}{R}$

(C) $N_B = \frac{GmM}{R^2} - m\omega^2 R - \frac{mv^2}{R}$

(D) $N_B = \frac{GmM}{R^2} + m\omega^2 R - \frac{mv^2}{R}$

8. A mass m of volume V = 10cm^2 is tied with a string to the base of a container filled with a liquid of density $\left(P_\ell = 10^3 kg/m^3\right)$ as shown in the figure. The container is then accelerated with acceleration (a_x = 9 m /s^2) and (a_y = 2 m /s^2) s shown in the figure. [$g = 10\left(\frac{m}{s^2}\right)$ acceleration due to gravity] The net buoyancy force acting on the mass is

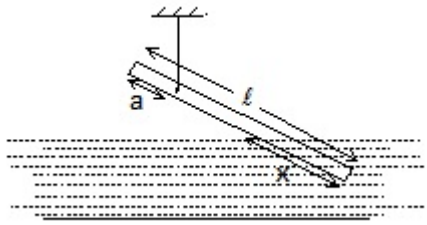


(A) 10N (B) 12N

(C) 15N (D) 9N

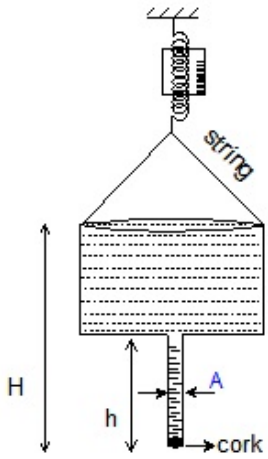
Space for Rough work

9. One end of a uniform stick having length ℓ = 1m is hung from a string as shown in figure. Length a = 20cm is above the point of suspension. The length of the stick below the surface of water is $\left[\rho_{stick} / \rho_{liquid} = 0.25\right]$



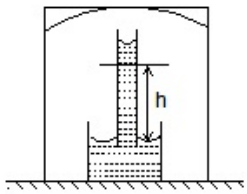
(A) x = 0.1 m
(B) x = 0.15 m
(C) x = 0.2 m
(D) x = 0.25 m

10. A vessel completely filled with water is suspended with help of strings from a spring balance as shown in figure. A cork is plugged at the bottom of the vessel. If the initial reading of the spring balance is W_0. Then the reading of the spring balance just after the cork is removed and the liquid begins to flow out will be [ρ = density of liquid & A = cross – sectional area of narrow pipe]



(A) $W_0 + \rho gHA$ (B) $W_0 + 2\rho gHA$
(C) $W_0 - \rho gHA$ (D) $W_0 - 2\rho gHA$

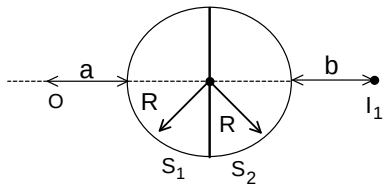
11. A capillary tube of inner radius r = 5mm is covered in a container with a liquid whose vapour pressure is negligible. The density of the liquid is ρ = 1000 kg / m^3. The container and the tube are in vacuum condition. The pressure inside liquid in the capillary tube at a height h = 20 cm as shown in figure is [g = 10 m / s^2]



(A) $+2000\,N/m^2$ (B) $-2000\,N/m^2$
(C) $\left(10^5 - 1000\right)\frac{N}{m^2}$ (D) $\left(10^5 - 2000\right)\frac{N}{m^2}$

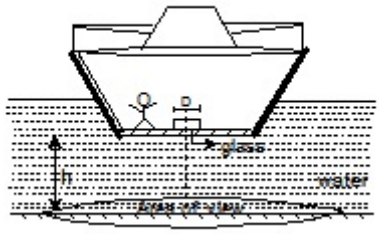
Space for Rough work

12. An object O kept a distance a = 10 cm from the surface of a glass hemisphere S_1 of radius R. S_2 is another glass hemisphere kept side by side of same radius R = 5cm as shown in figure. I_1 is the image formed due to object O (b = 6cm). Now, if S_2 is removed and S_1 is kept at the same place and its flat surface is silvered. Then find the distance I_1I_2. (where I_2 is image formed by silvered S_1 only). (μ glass = 3/2)



(A) 10 cm (B) 12 cm
(C) 11 cm (D) 22 cm

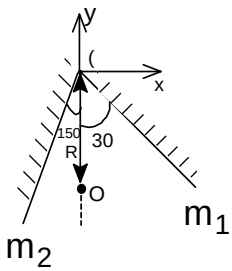
13. A glass slab of diameter D = 20 cm is made at the base of a ship as shown in figure. Diameter of glass slab is much larger than the thickness of the glass slab. The area 'S' of the field of view at the sea bottom if $\mu_{water} = 4/3$, $\mu_{glass} = 3/2$ & the depth of sea base below the slab is $h = \sqrt{7}m$ is



(A) $9\pi\ m^2$ (B) $\left(\frac{88}{5}\right)m^2$

(C) $\frac{1408}{17}m^2$ (D) $\frac{\pi}{100}m^2$

14. Two plane mirrors intersect at origin. An object 'O' is kept at a distance R on negative y axis as shown in the figure. The co – ordinates of images formed after three reflections will be (first reflection is considered at m_1)



(A) $\left(-R/2,+\sqrt{3}R/2\right)(-R,0)\ \&\ \left(+\sqrt{3}R/2,+R/2\right)$

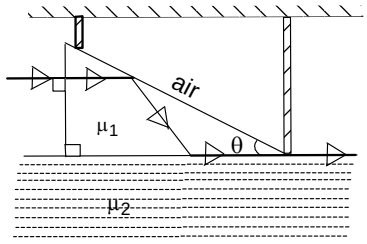
(B) $\left(-\sqrt{3}R/2,+R/2\right)(+R,0)\ \&\ \left(+\frac{R}{\sqrt{2}},+\frac{R}{\sqrt{2}}\right)$

(C) $\left(\sqrt{3}R/2,-R/2\right)(-R,0)\ \&\ \left(+R/2,+\sqrt{3}R/2\right)$

(D) $\left(\frac{-R}{\sqrt{2}},\frac{R}{\sqrt{2}}\right)(+R,0)\ \&\ \left(+\frac{R}{2},+\sqrt{3}\frac{R}{2}\right)$

Space for Rough work

15. A right angled triangular prism of refractive index μ_1 is kept floating on a liquid (surface) having refractive index μ_2. The condition such that light incident normally passes through base of prism as shown in figure.



(A) $\mu_1 = \text{Cos}2\theta / \text{Cos}\theta$

(B) $\mu_2 = \text{Cos}2\theta / \text{Cos}\theta$

(C) $\mu_1 = \text{Sin}2\theta / \text{Cos}\theta$

(D) $\mu_2 = \text{Sin}2\theta / \text{Sin}\theta$

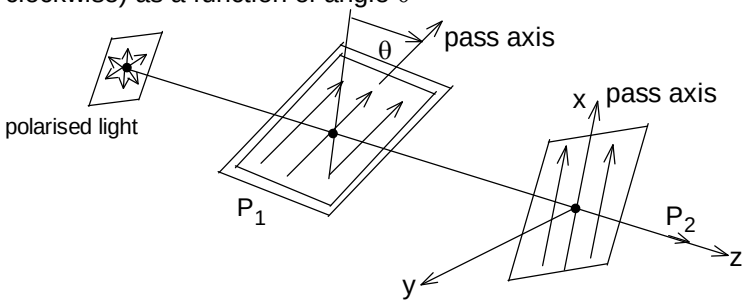
16. If $E = E_0 \sin(kz - \omega t)$ and $B = B_0 (kz - \omega t)$ are electric and magnetic field produced by an electromagnetic wave travelling in +z direction in a medium. Then if $\eta = \frac{E_0}{B_0}$, then the value of η is

[μ = permeability of a medium, ε = permittivity of medium]

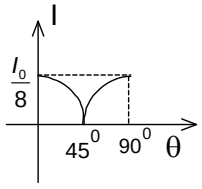
(A) $\sqrt{\in}$ /H (B) $\sqrt{\mu / \in}$

(C) $\sqrt{\mu \in}$ (D) $\frac{1}{\sqrt{\mu \in}}$

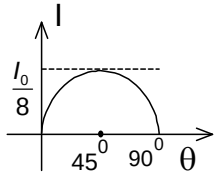
17. If an polarised light of intensity $I_0/2$ is passed through two polariods P_1 and P_2 as shown in figure. Then the intensity of light observed by an observer on other side of P_2 (if the polaroid P_1 is rotated clockwise) as a function of angle θ



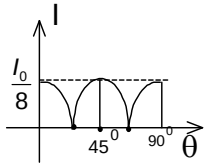
(A)



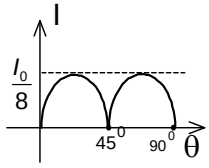
(B)



(C)

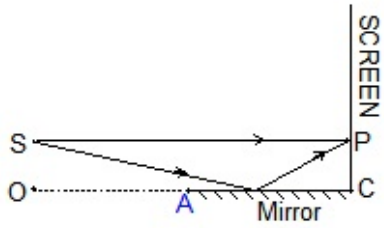


(D)



Space for Rough work

18. In a Lyod's mirror set up for interference of waves of frequency $6\times10^{14}s^{-1}$ emitted from point source S. Distance SO, OA & AC are 1mm, 50 cm and 50 cm respectively, Then the number of dark and bright fringes obtained on the screen will be

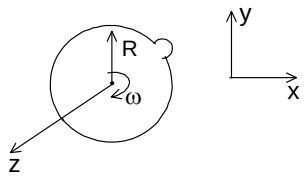


(A) Dark = 4 , Bright = 4
(B) Dark = 4 , Bright = 5
(C) Dark = 5 , Bright = 4
(D) Dark = 5 , Bright = 5

19. A point source of sound is vibrating with a frequency of 256 vibrations per second in air and propogating energy uniformly in all directions at the rate of 5 joules per second. Calculate the amplitude of the wave at a distance of 25 m from the source. Assume non–dissipative medium [c = 330 m / s & density of air = 1.29 kg / m^3]

(A) 0.8×10^{-2}m (B) 0.8×10^{-8}m
(C) 10^{-6}m (D) 10^{-4}m

20. A small pulse is generated on a ring rotating with a constant angular velocity ω as shown in the figure. Then the speed of pulse with respect to ground



(A) ωR (B) $\omega R/2$
(C) $2\omega R$ (D) zero

21. A sound detector is moving with a constant velocity towards a stationary sound source. The source emits a beep of sound of frequency f = 262 Hz that propogates with speed c = 330 m/s. The detector detects the beep of sound over a distance L = 78 m and registers a frequency f = 275 Hz. Find duration of the beep emitted by the source.

(A) 2s (B) 4s
(C) 5s (D) 6s

22. Two boats are anchored a distance 'd' apart in a sea, where seabed is at a uniform depth of D and consists of a flat rock bed. If a gun is fired at a boat, in how much time will the firing be heard at the other boat? (Given: speed of sound in air is v_a, in the water is v_w and in the rock is v_r)

(A) d / v_a (B) $2D / v_w + d / v_r$
(C) $\dfrac{dv_w + 2D\sqrt{v_r^2 - v_w^2}}{v_r v_w}$ (D) $\dfrac{d}{v_r} + \left(\dfrac{2Dv_r}{\sqrt{v_r^2 - v_w^2}}\right)\times\dfrac{1}{v_w}$

Space for Rough work

23. If photoelectrons emitted by a metal on irradiating it by a light of wave length λ cannot move away farther than a distance 'd' in presence of a retarding electric field E, find threshold wavelength of the metal

(A) $\lambda c / eEd$ (B) $\lambda + \frac{hc}{eEd}$

(C) $\lambda - \frac{hc}{eEd}$ (D) $\left[\frac{1}{\lambda} - \frac{eEd}{hc}\right]^{-1}$

24. A narrow beam of monochromatic light of wavelength λ emitted from a source of power P is propogating in the positive x–direction. After being reflected from a perfectly reflecting plane mirror of area vector $\vec{A} = A\left(-\hat{i} - \hat{j}\right)$, the beam falls on a metal plate of surface area A [$\vec{A}$ is along outward normal at mirror]. The force exerted by light beam on the mirror is $\vec{F}$ and $\Delta\vec{p}$ is change in momentum of each photon then[c is speed of light in vaccuum]

(A) $\Delta\vec{F} = \frac{P}{c}\left(\hat{i} + \hat{j}\right), \Delta\vec{p} = \frac{h}{\lambda}\left(-\hat{i} - \hat{j}\right)$ (B) $\Delta\vec{F} = \frac{P}{c}\left(\hat{i} + \hat{j}\right), \Delta\vec{p} = \frac{h}{\lambda}\left(\hat{i} + \hat{j}\right)$

(C) $\Delta\vec{F} = \frac{P}{c}\left(-\hat{i} - \hat{j}\right), \Delta\vec{p} = \frac{h}{\lambda}\left(-\hat{i} - \hat{j}\right)$ (D) $\Delta\vec{F} = \frac{P}{c}\left(-\hat{i} - \hat{j}\right), \Delta\vec{p} = \frac{h}{\lambda}\left(+\hat{i} + \hat{j}\right)$

25. A hydrogen molecule in excited state travelling in the x–direction with kinetic energy of 1.0 eV dissociate into two hydrogen atoms. If one of them moves perpendicular to the x–direction with a kinetic energy 1eV, then the energy released in the dissociation reaction will be

(A) 2.6 eV (B) 2eV

(C) 3.6 eV (D) 3 eV

26. Two resistors of resistance $R_1 = (6 \pm 0.09)\Omega$ and $R_2 = (3 \pm 0.09)\Omega$ are connected in parallel. The equivalent resistance R with error (in Ω)

(A) $R = (2 \pm 0.04)$ (B) $R = (2 \pm 0.05)$

(C) $R = (9 \pm 0.18)$ (D) $R = (2 \pm 0.18)$

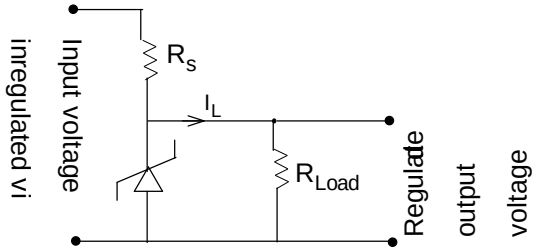
Space for Rough work

27. The diameter of a solid sphere is measured by vernier calipers (10 VSD = 9 MSD) and (1 MSD = 1 mm) the reading of main scale is 20 & Vernier scale reading is 1. The volume of sphere will be

(A) $4.23\,cm^3$ (B) $4.25\,cm^3$

(C) $4.232\,cm^3$ (D) $4.253\,cm^3$

28. In a zener diode (dc voltage regulator) circuit with V_{zener} = 7.0 V is used for regulation. The load current is to be 4.0 mA and the unregulated input is 11.0 V. What should be the value of series resistor R_s if zener diode current is five times the load current



(A) 120Ω (B) 167Ω

(C) 180Ω (D) 200Ω

29. Photo diodes are preferable used in reverse bias condition for measuring light intensity because

(A) Current in reverse bias is less which is easier to measure compared to higher currents in forward bias

(B) High fractional change due to photo effects in minority carrier dominated reverse bias current is more easily measurable than the high fractional change in the forward bias current

(C) Low fractional change due to photo–effects in minority carries dominated reverse bias current is more easily measurable than the high fractional change in the forward bias current

(D) Current is zero in reverse bias set–up

30. A transmitting antenna at the top of a tower has a height 64m and the height of the receiving antenna is 100 m. The maximum distance between them for satisfactory communication in LOS mode will be {approximately}

(A) 45.5 km (B) 91 km

(C) 136.5 km (D) 182.0 km

Space for Rough work

Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. Which of the following statements about photochemical smog is wrong?
(A) it has high concentration of oxidizing agents
(B) it has low concentration of oxidizing agents
(C) it can be controlled by controlling the release of NO_2, hydrocarbons, ozone etc
(D) plantation of some plants like pinus helps in controlling photochemical smog

32. Compound which is added to soap to impart antiseptic properties is
(A) sodium laurylsulphate
(B) sodium dodecylbenzenesulphonate
(C) Rosin
(D) Bithional

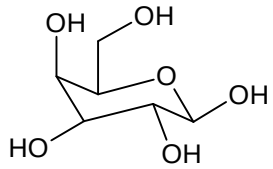
33. 1.0 mole of Fe reacts completely with 0.65 mole of O_2 to give a mixture of only FeO and Fe_2O_3. The mole ratio of ferrous oxide to ferric oxide is
(A) 2 : 2
(B) 4 : 3
(C) 1 : 2
(D) 2 : 7

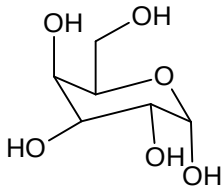
34. Gold number of haemoglobin is 0.03. Hence, 100 ml of gold sol will require haemoglobin so that gold is not coagulated by 10 ml of 10% NaCl solution
(A) 0.03 mg
(B) 30 mg
(C) 0.30 mg
(D) 3 mg

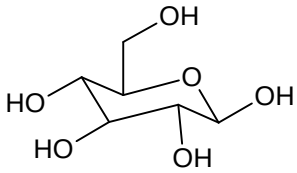
35. Which of the following statement is not true regarding rayon?
(A) it is pure regenerated cellulose
(B) it is obtained by dissolving wood pulp in alkaline CS_2
(C) it is obtained by passing Na – salt of cellulose Xanthate through spinneret into aqueous $NaHCO_3$ solution
(D) it is extracted as fibres of cellulose

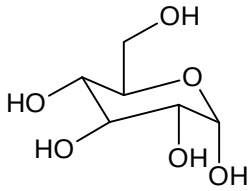
Space for Rough work

36. Galactose is epimeric with glucose at C_4. Which of the following structures represents β-D-galactopyranose?

(A) 

(B) 

(C) 

(D) 

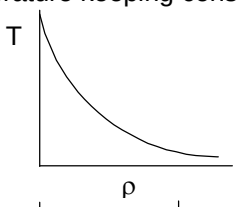
37. Iodate ions can be reduced to iodine by iodide ions as per the given redox reaction in acidic medium

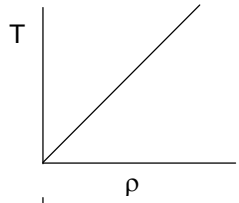
$IO_3^-(aq) + I^-(aq) \longrightarrow I_2(S)$.

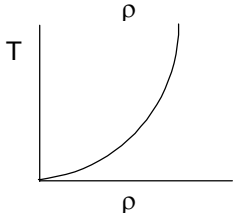
How many moles of iodine are produced for every mole of iodate ions consumed in the reaction?

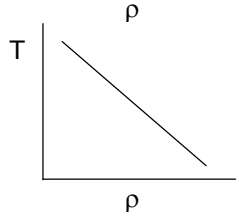
(A) 1 (B) 2.5
(C) 3 (D) 5

38. Which of the following is a graph of the relation of the density (ρ) of a gas sample to the absolute temperature keeping constant pressure

(A) 

(B) 

(C) 

(D) 

Space for Rough work

39. If m_p and m_α are masses of proton and α-particles, find ratio of momentum of a proton and an α-particle which are accelerated from rest by a potential difference of 200 V

(A) $\frac{m_p}{2m_\alpha}$ (B) $\frac{2m_\alpha}{m_p}$

(C) $\sqrt{\frac{m_p}{2m_\alpha}}$ (D) $\sqrt{\frac{2m_p}{m_\alpha}}$

40. Consider the cell $Ag\left|AgBr(s)Br^-\right|\left|AgCl(s)Cl^-\right|Ag$ at 298 K. The solubility product of AgCl and AgBr are 1×10^{-10} and 5×10^{-13} respectively. What should be the ratio of concentration of Br^- and Cl^- by which emf of the cell becomes zero?

(A) 200 (B) $\frac{1}{200}$

(C) $\frac{1}{300}$ (D) none of these

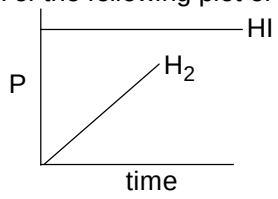
41. Calculate the entropy change for vaporization of water if latent heat of vaporization for water is 2.26 kJ/gram. The K_b for H_2O is 0.51 K/molality.

(A) 26.33 cal/mole (B) 23.33 cal/mole

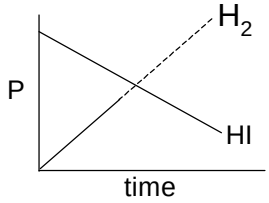
(C) 1 KJ/mole (D) 30 cal/mole

42. Which of the following plot of pressure (P) Vs time satisfy decomposition of HI on gold surface

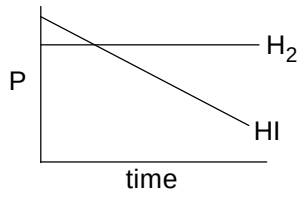
(A)



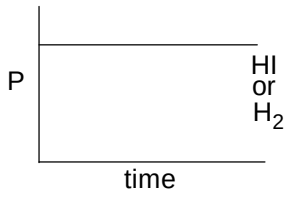
(B)



(C)

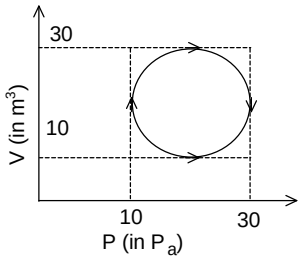


(D)



Space for Rough work

43. A system works under cyclic process as follows



Heat absorbed during the process is

(A) $\frac{22}{7}\times10^2\ J$ (B) $\frac{22}{7}\times10^3\ J$

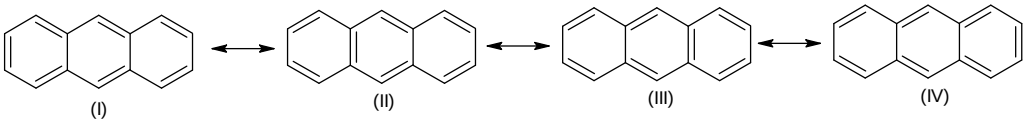
(C) $\frac{22}{7}\times10^4\ J$ (D) $\frac{22}{7}\times10^5\ J$

44. Which of the following statement is correct about the cyclic silicate $[Si_6O_{18}]^{n-}$?
(A) the value of n is 12
(B) each Si atom is bonded with three oxygen atoms
(C) each oxygen atom is bonded with two Si atoms
(D) all of these

45. Ores of calcium are
(A) gypsum, colmanite, hydrolith (B) quick lime, limestone, pearl ash
(C) quick lime, fluorapatite, feldspar (D) carnallite, pearl ash, feldspar

46. Find the incorrect statement
(A) $TiH_2 > CaH_2 > BeH_2$, order of electrical conductance
(B) CsH > NaH > LiH, order of ionic character of hydrides
(C) $H_2 > D_2 > F_2$, order of bond dissociation enthalpy
(D) $NaH > MgH_2 > H_2O$, order of reducing character

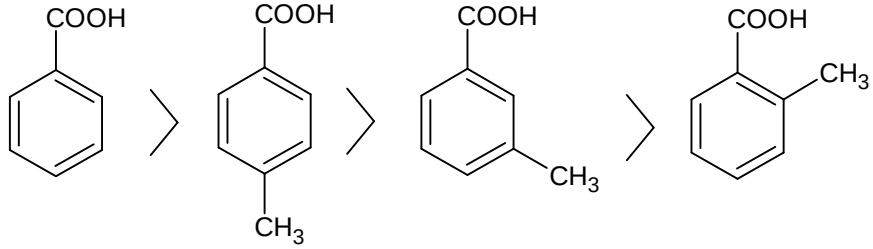
47. The correct stability order of given resonating structure is

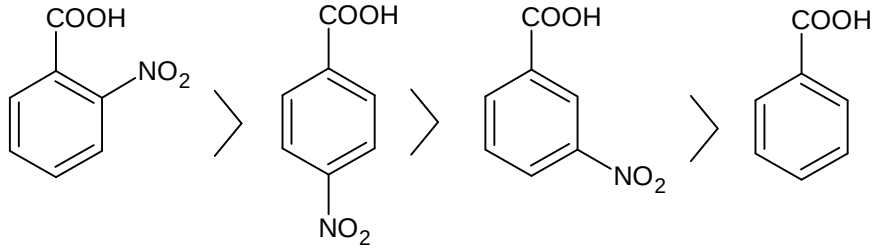


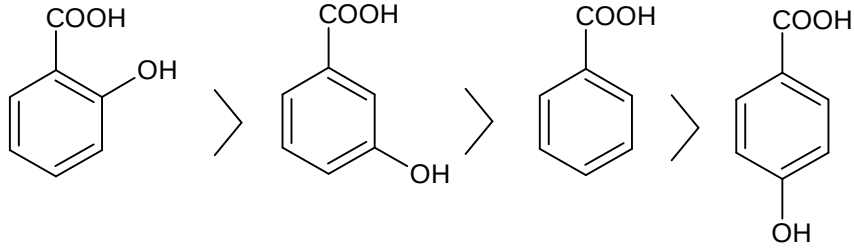
(A) I = II = III = IV (B) IV > III > II > I
(C) I > II > III > IV (D) II = III > I = IV

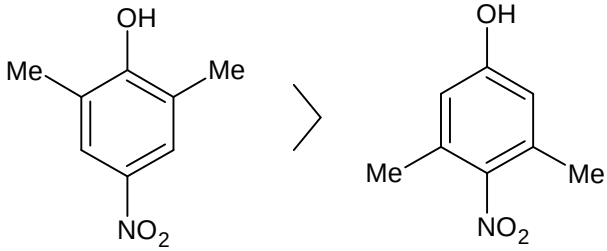
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48. Find the incorrect order of acidic strength of following

(A) 

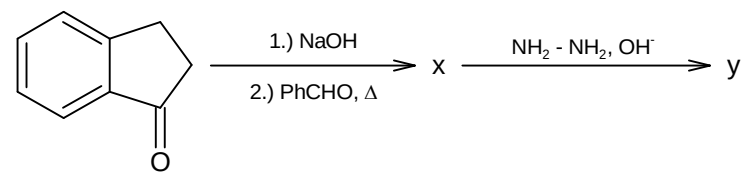
(B) 

(C) 

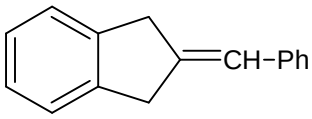
(D) 

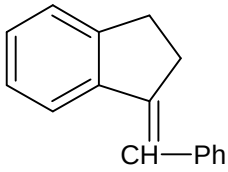
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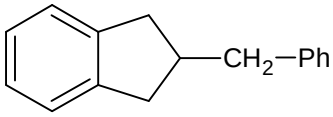
49.

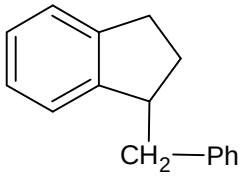


Product y is

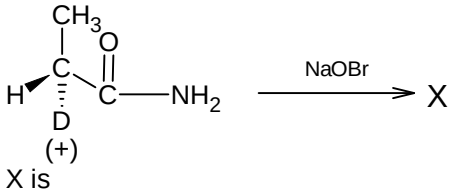
(A) 

(B) 

(C) 

(D) 

50.



X is

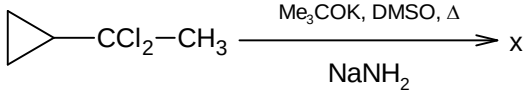
(A) $(-)CH_3 - CHD - NH_2$

(B) $(+)CH_3 - CHD - NH_2$

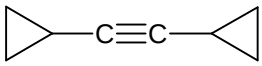
(C) $(\pm)CH_3 - CHD - NH_2$

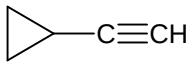
(D) $(+)\overset{Br}{\underset{|}{C}}H_3 - CHD - NH_2$

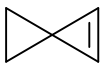
51.

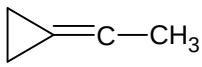


Product x is

(A) 

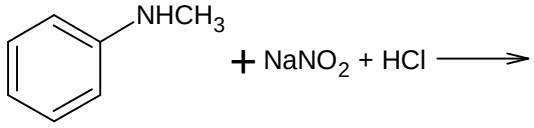
(B) 

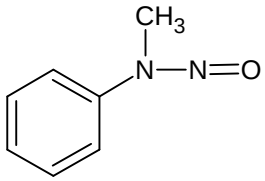
(C) 

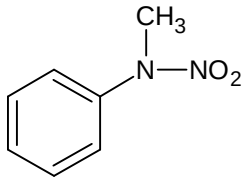
(D) 

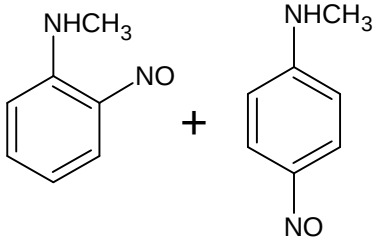
Space for Rough work

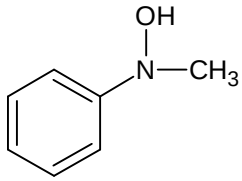
52. Predict the product

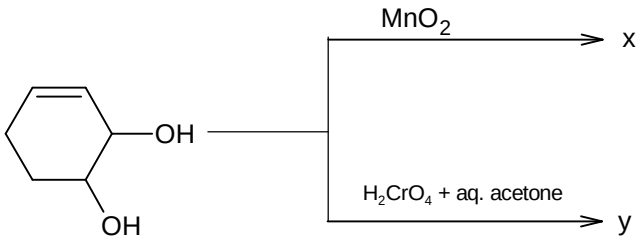


(A) 

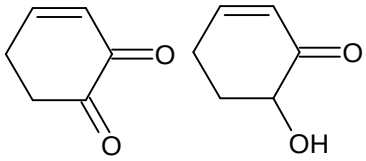
(B) 

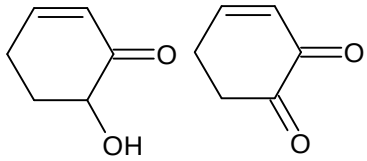
(C) 

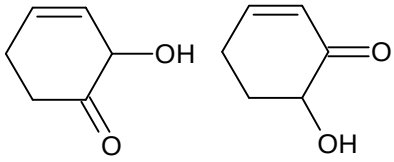
(D) 

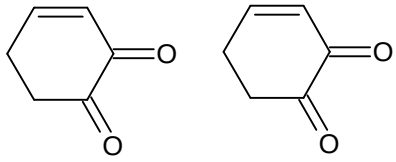
53. 

The compounds x and y are

(A) 

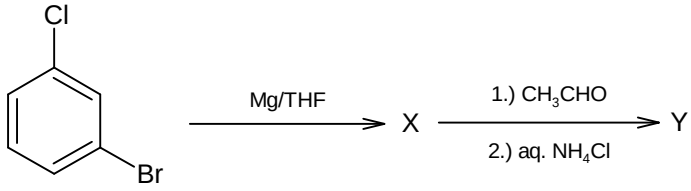
(B) 

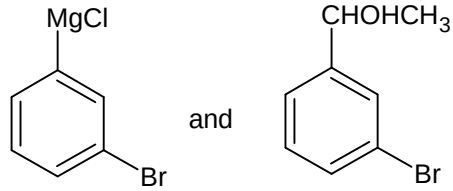
(C) 

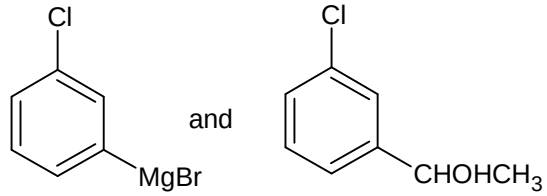
(D) 

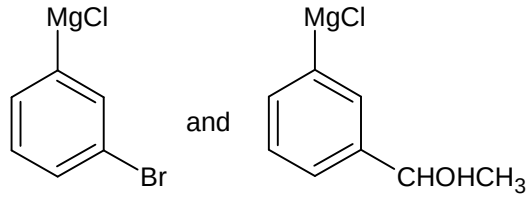
Space for Rough work

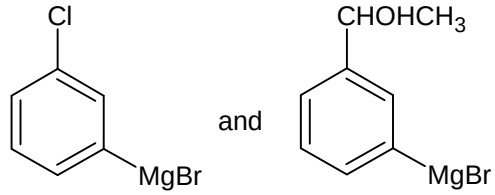
54. What are x and y in the following reaction?



(A) 

(B) 

(C) 

(D) 

55. An acid base indicator has $K_a = 1.0 \times 10^{-5}$. The acid form of the indicator is red and the basic form is blue. Calculate the pH change required to change the colour of the indicator from 80% red to 80% blue.

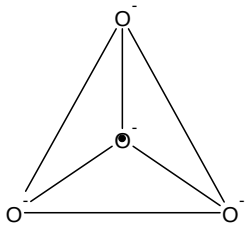
(A) 1.20 (B) 0.80
(C) 0.20 (D) 1.40

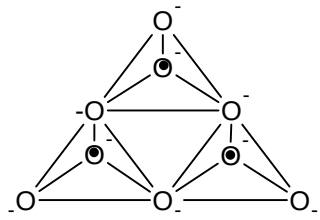
Space for Rough work

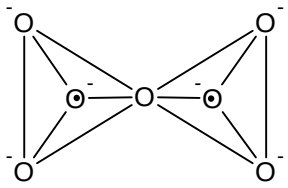
56. For 8 : 8 coordinated CsCl structure, how many Cs^+ ions occupy the second-nearest neighbour locations of a Cs^+ ion?
(A) 8 (B) 24
(C) 6 (D) 12

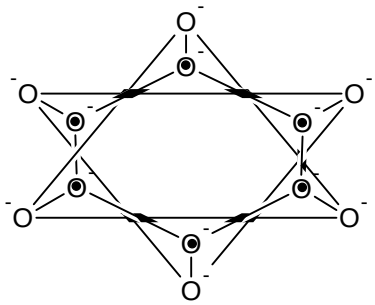
57. CO_3^{2-} and SO_3^{2-} can be distinguished from each other by the production of CO_2 and SO_2 respectively upon using
(A) Baryta water (B) Lime water
(C) acidified dichromate (D) sulphamic acid

58. Which of the following represents a pyrosilicate structure?

(A) 

(B) 

(C) 

(D) 

59. Nessler's reagent is used to detect NH_3. This reagent is prepared by mixing
(A) KI (excess) + $HgCl_2$ + KOH (B) KCl + HgI_2 + KOH
(C) KI (excess) + $HgCl_2$ + $NH_4^+OH^-$ (D) KI (excess) + Hg_2Cl_2 + I_2 solution

60. Ferrocene is a compound of
(A) one Fe^{2+} and two $C_5H_5^-$ ions (B) Two Fe^{2+} and two $C_5H_5^-$ ions
(C) one Fe^{3+} and two $C_5H_5^-$ ions (D) Two Fe^{3+} and six $C_5H_5^-$ ions

Space for Rough work

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. The range of the function $f(x) = \sqrt{4-x^2} + \sqrt{x^2-1}$ is

(A) $\left[\sqrt{3}, \sqrt{7}\right]$ (B) $\left[\sqrt{3}, \sqrt{5}\right]$

(C) $\left[\sqrt{2}, \sqrt{3}\right]$ (D) $\left[\sqrt{3}, \sqrt{6}\right]$

62. If the variable line $y = kx + 2h$ is tangent to an ellipse $2x^2 + 3y^2 = 6$, then locus of $P(h, k)$ is a conic C whose eccentricity equals

(A) $\frac{\sqrt{5}}{2}$ (B) $\sqrt{\frac{7}{3}}$

(C) $\frac{\sqrt{7}}{3}$ (D) $\sqrt{2}$

63. The number of solution(s) of the equation $z^2 = 4z + |z|^2 + \frac{16}{|z|^3}$ is

(A) 0 (B) 1

(C) 2 (D) 3

64. The area of the region bounded by the curve C : $y = \frac{x+1}{x^2+1}$ and the line L : $y = 1$, is

(A) $1 - \frac{1}{2}\ln 2 + \frac{\pi}{4}$ (B) $\ln 2 - \frac{\pi}{4} + 1$

(C) $\frac{1}{2}\ln 2 + \frac{\pi}{4} - 1$ (D) $\ln 2 - \frac{\pi}{2} + 1$

Space for Rough work

65. The solution set of inequality $(\cot^{-1}x)(\tan^{-1}x) + \left(2-\frac{\pi}{2}\right)\cot^{-1}x - 3\tan^{-1}x - 3\left(2-\frac{\pi}{2}\right) > 0$, is

(A) $x \in (\tan 2, \tan 3)$ (B) $x \in (\cot 3, \cot 2)$
(C) $x \in (-\infty, \tan 2) \cup (\tan 3, \infty)$ (D) $x \in (-\infty, \cot 3) \cup (\cot 2, \infty)$

66. Number of solutions of the equation $\tan x + \sec x = 2\cos x$ in $(-\pi, 2\pi)$, is
(A) 1 (B) 2
(C) 3 (D) 4

67. Inside the unit circle $S = \{(x, y) \mid x^2 + y^2 = 1\}$ there are three smaller circles of radius equals to a, tangent to each other and to S. The value of a equals

(A) $\sqrt{2}\left(\sqrt{2}-1\right)$ (B) $\sqrt{3}\left(2-\sqrt{3}\right)$
(C) $\sqrt{2}\left(2-\sqrt{3}\right)$ (D) $\sqrt{2}\left(2-\sqrt{3}\right)$

68. If [x] and {x} denotes respectively, the greatest integer function less than or equal to x and fractional part function, then the number of real x, satisfying the equation $(x-2)[x] = \{x\} - 1$, is
(A) 0 (B) 1
(C) 2 (D) infinite

69. If $g(x) = \int_1^x e^{t^2}\,dt$ then the value of $\int_3^{x^3} e^{t^2}\,dt$ equals

(A) $g(x^3) - g(3)$ (B) $g(x^3) + g(3)$
(C) $g(x^3) - 3$ (D) $g(x^3) - 3g(x)$

70. For $\theta \in \left(0, \frac{\pi}{2}\right)$, the value of definite integral $\int_0^{\theta} \ln(1+\tan\theta\tan x)\,dx$ is equal to

(A) $\theta\, \ln(\sec\theta)$ (B) $\theta\, \ln(\operatorname{cosec}\theta)$
(C) $\frac{\theta \ln 2}{2}$ (D) $2\theta\, \ln \sec\theta$

71. If r, s are the roots of $a_0 + a_1x + a_2x^2 = 0$ ($a_0, a_1, a_2 \in R$ and $a_2 \neq 0$), then the equality $a_0 + a_1x + a_2x^2 = a_0\left(1-\frac{x}{r}\right)\left(1-\frac{x}{s}\right)$ holds

(A) for all values of x, $a_0 \neq 0$ (B) only when x = 0
(C) only when x = r or x = s (D) only when x = r or x = s, $a_0 \neq 0$

Space for Rough work

72. The position vectors of vertices of ΔABC are (1, –2), (– 7, 6) and $\left(\frac{11}{5},\frac{2}{5}\right)$ respectively. The measure of the interior angle A of the triangle is
(A) acute and lies in (75°, 90°) (B) acute and lies in (60°, 75°)
(C) acute and lies in (45°, 60°) (D) obtuse

73. Consider the graphs of $y = Ax^2$ and $y^2 + 3 = x^2 + 4y$, where A is a positive constant and $x, y \in R$. Number of points in which the two graphs intersect, is
(A) exactly 4
(B) exactly 2
(C) at least 2 but the number of points varies for different positive values of A.
(D) zero for atleast one positive A.

74. If $a^2 + b^2 + c^2 + ab + bc + ca \le 0$, $a, b, c \in R$ then value of the determinant
$$\begin{vmatrix} (a+b+2)^2 & a^2+b^2 & 1 \\ 1 & (b+c+2)^2 & b^2+c^2 \\ c^2+a^2 & 1 & (c+a+2)^2 \end{vmatrix}$$
(A) 65 (B) $a^2 + b^2 + c^2 + 31$
(C) $4(a^2 + b^2 + c^2)$ (D) 0

75. Let $f'(x) = e^{x^2}$ and $f(0) = 10$ and $A < f(1) < B$ can be concluded from the Mean Value Theorem then largest value of (A – B) equals
(A) e (B) 1 – e
(C) e – 1 (D) 1 + e

76. Water is dropped at the rate of $2m^3$/sec. into a cone of semi-vertical angle 45°. The rate at which periphery of water surface changes when height of the water in the cone is 2 meter, is
(A) 2 m/sec. (B) 1 m/sec.
(C) 3m/sec. (D) 4 m/sec.

77. Let S denote sum of the series $\frac{3}{2^3}+\frac{4}{2^4\cdot 3}+\frac{5}{2^6\cdot 3}+\frac{6}{2^7\cdot 5}+.....\infty$. Compute the value of S^{-1}.
(A) 8 (B) 6
(C) 4 (D) 2

Space for Rough work

78. Two loaded dice each have the property that 2 or 4 is three times as likely to appear as 1, 3, 5 or 6 on each roll. When two such dice are rolled, the probability of obtaining a total of 7, is

(A) $\frac{1}{8}$ (B) $\frac{1}{7}$

(C) $\frac{7}{50}$ (D) $\frac{7}{25}$

79. A line passes through the point $A\left(\hat{i}+2\hat{j}+3\hat{k}\right)$ and is parallel to the vector $\vec{V}=\left(\hat{i}+\hat{j}+\hat{k}\right)$. The shortest distance from the origin, of the line is

(A) $\sqrt{2}$ (B) $\sqrt{4}$

(C) $\sqrt{5}$ (D) $\sqrt{6}$

80. If z is a complex number satisfying the equation $\left| z - (1 + i) \right|^2 = 2$ and $\omega = \frac{2}{Z}$, then the locus traced by 'ω' in the complex plane is

(A) x – y – 1 = 0 (B) x + y – 1 = 0

(C) x – y + 1 = 0 (D) x + y + 1 = 0

81. Let a and b be positive numbers not equal to 1 and $\frac{3\pi}{2}<\theta<2\pi$. If $a^{\tan\theta} > b^{\tan\theta} > 1$, then which one of the following is always **TRUE**?

(A) a > b > 1 (B) a < b < 1

(C) b < a < 1 (D) b > a > 1

82. If the equation $(x^2 + a|x| + a + 1)(x^2 + (a + 1)|x| + a) = 0$ has no real root, then the range of values of 'a' is

(A) (0, 1) (B) $(-\infty, 0)$

(C) $(0, \infty)$ (D) (–1, 0)

83. The area of the triangular region in the first quadrant bounded on the right by the y-axis, bounded above by the line 7x + 4y = 168 and bounded below by the line 5x + 3y = 121, is

(A) $\frac{50}{3}$ (B) $\frac{52}{3}$

(C) $\frac{53}{3}$ (D) 17

Space for Rough work

84. If p_1, p_2 are the roots of the quadratic equation $ax^2 + bx + c = 0$ and q_1, q_2 are the roots of the quadratic equation $cx^2 + bx + a = 0$ (a, b, c $\in$ R) such that p_1, q_1, p_2, q_2 are in A.P. of distinct terms, then a + c -1 equals

(A) – 1 (B) 1

(C) $\frac{1}{2}$ (D) 2

85. The locus of the middle point of the chord of the circle $x^2 + y^2 = 1$ such that the segment of the chord on the parabola $y = x^2 – x$ subtends a right angle at the origin, is a circle whose centre and radius respectively are

(A) (1, 1) and $\sqrt{2}$ (B) (1, 1) and 2

(C) $\left(\frac{1}{2},\frac{1}{2}\right)$ and $\frac{1}{2}$ (D) $\left(\frac{1}{2},\frac{1}{2}\right)$ and $\frac{1}{\sqrt{2}}$

86. If the roots of the equation $x^2 + 6x + l = 0$ are real and distinct and they differ by atmost 4, then the sum of all possible integral value(s) of l, is

(A) 16 (B) 26

(C) 36 (D) 35

87. A convex polygon has 44 diagonals. The polygon is

(A) 9 sided (B) 10 sided

(C) 11 sided (D) 12 sided

88. A class has three teachers, Mr. P, Ms. Q and Mrs. R and six students A,B,C,D,E,F. Number of ways in which they can be seated in a line of 9 chairs, if between any two teachers there are exactly two students, is k(6!) then the value of k is

(A) 18 (B) 12

(C) 24 (D) 6

89. Sixteen players S_1, S_2, S_3,..... , S_{16} play in a tournament. Number of ways in which they can be grouped into eight pairs so that S_1 and S_2 are in different groups, is equal to

(A) $\frac{(14)!}{2^6 \cdot 6!}$ (B) $\frac{(15)!}{2^7 \cdot 7!}$

(C) $\frac{(14)!}{2^7 \cdot 6!}$ (D) $\frac{(14)!}{2^6 \cdot 7!}$

90. Find the minimum value of the expression $E = |z|^2 + |z – 3|^2 + |z – 6i|^2$
(where $z = x + iy$, $x, y \in R$).

(A) 10 (B) 20

(C) 30 (D) 40

Space for Rough work